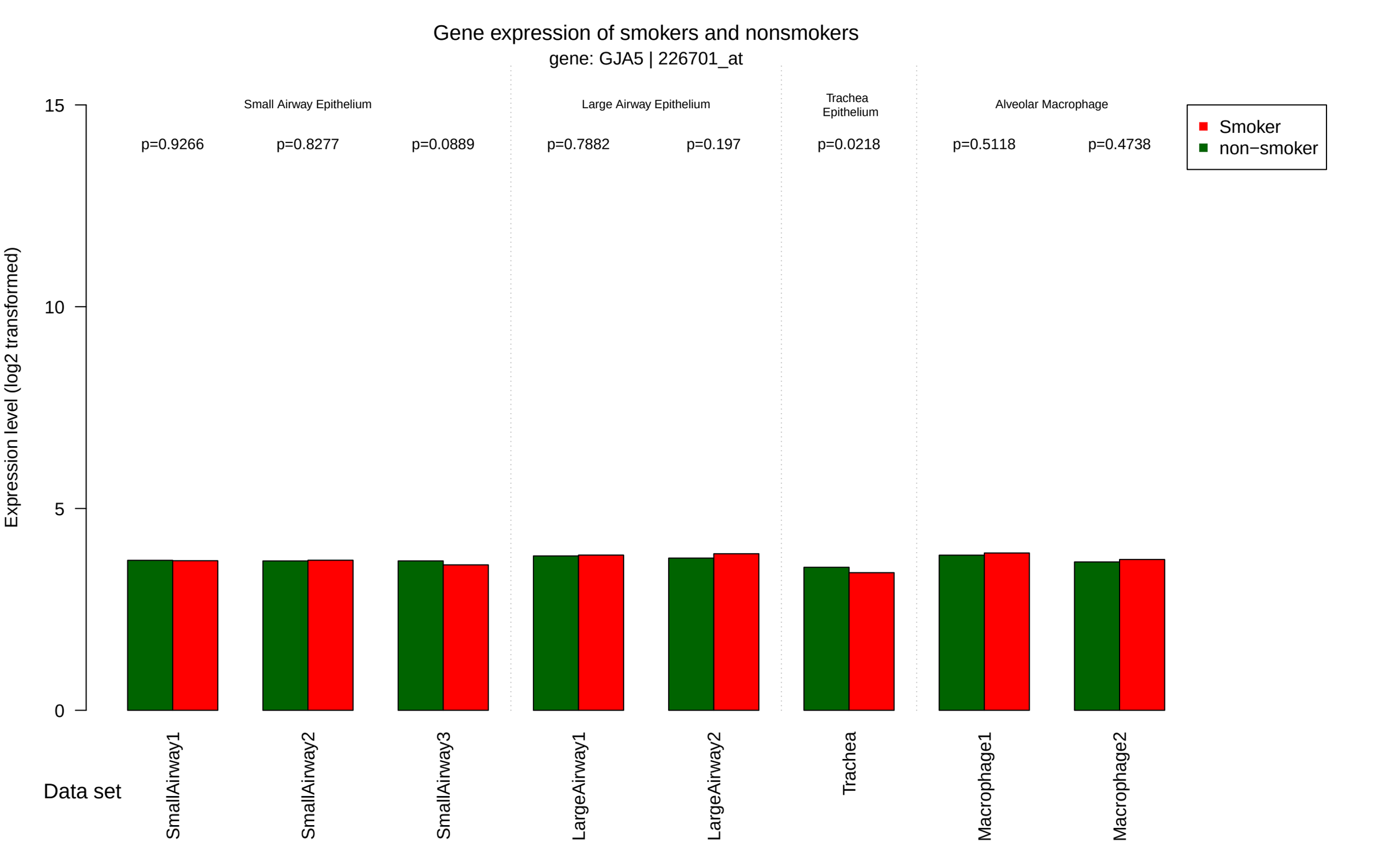
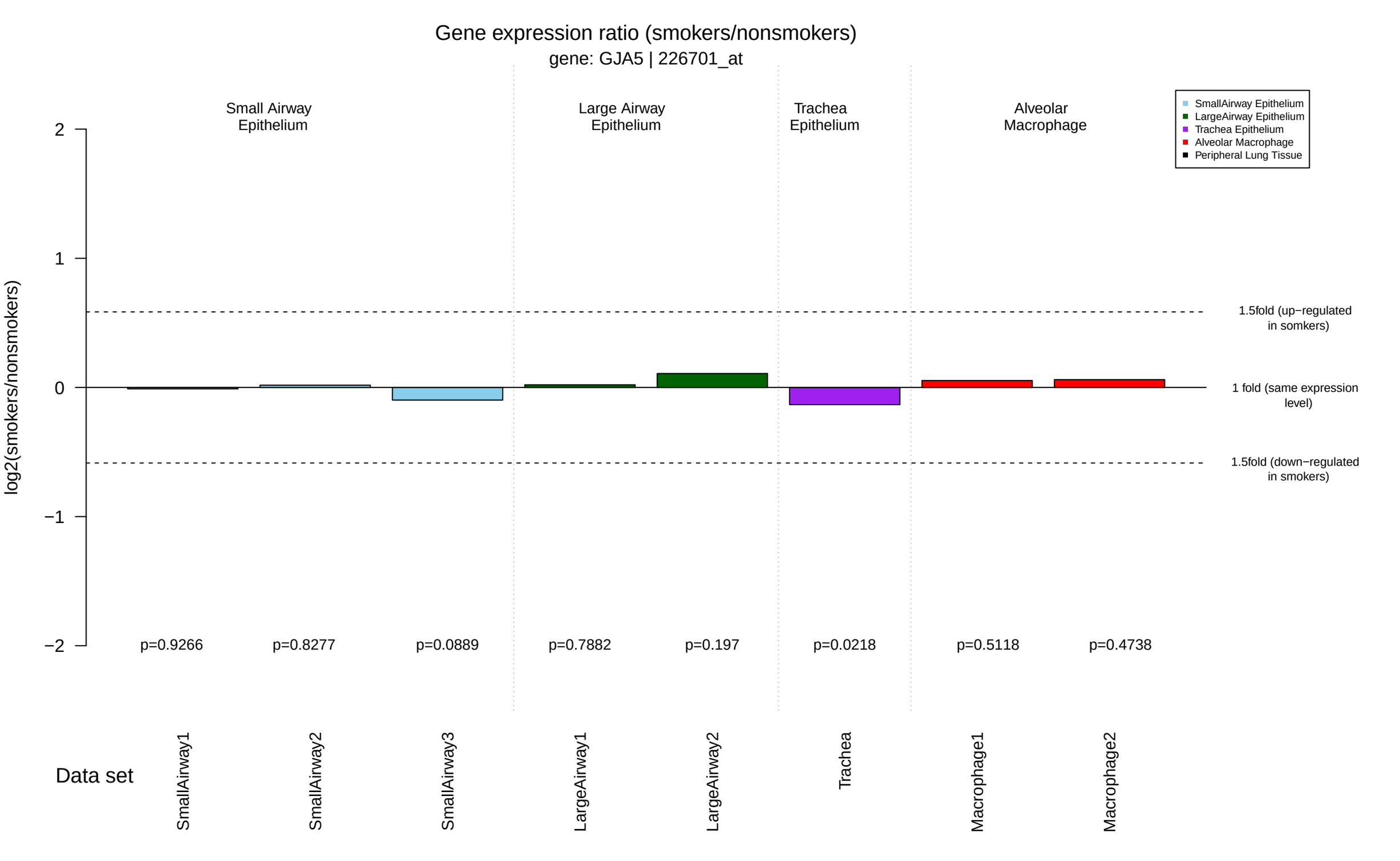
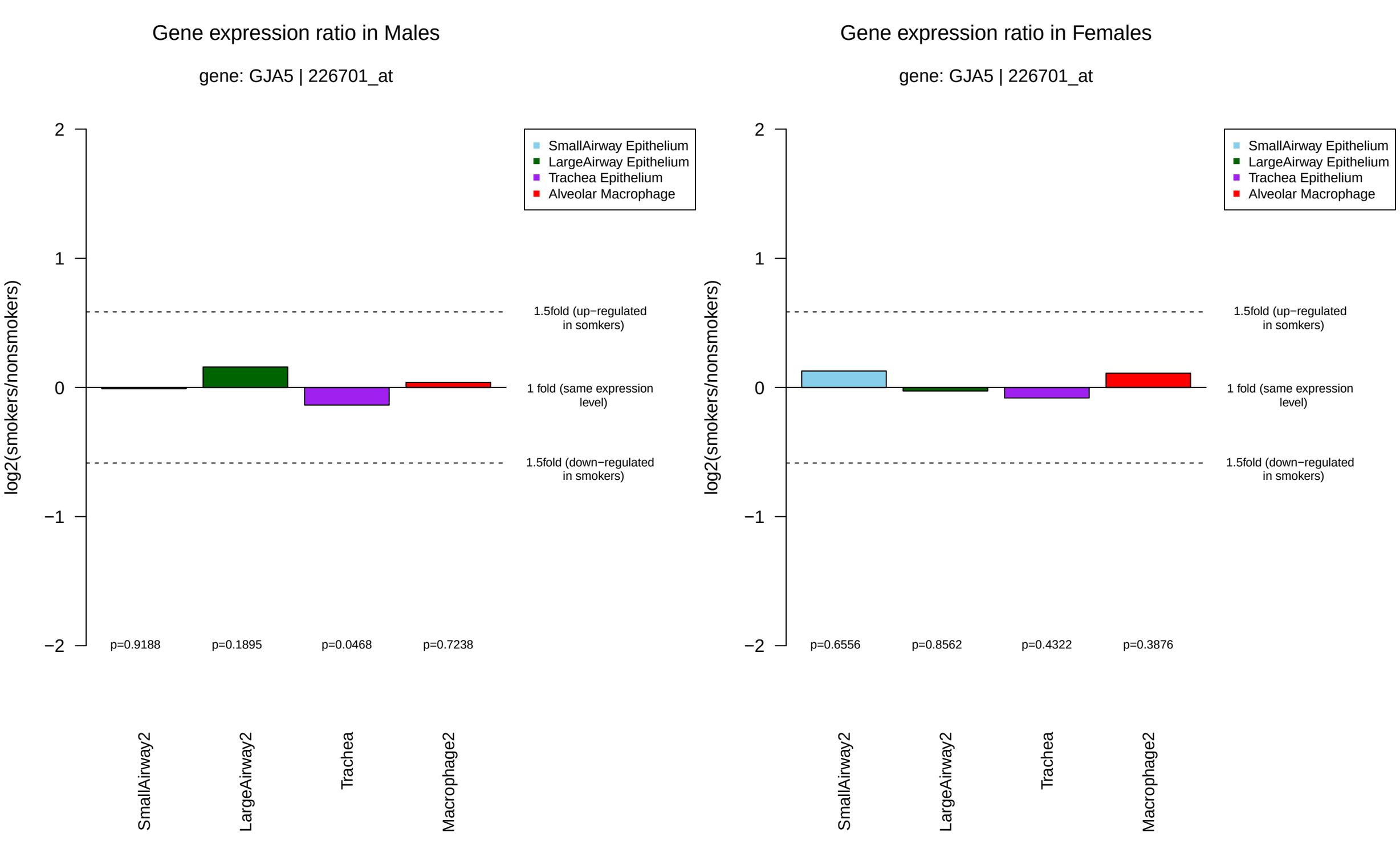
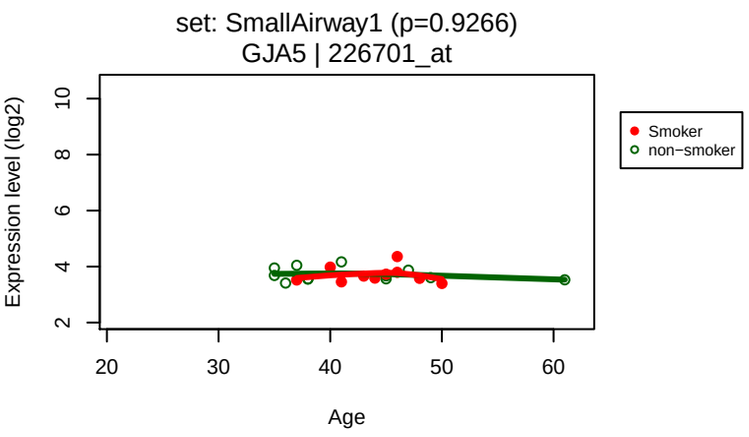
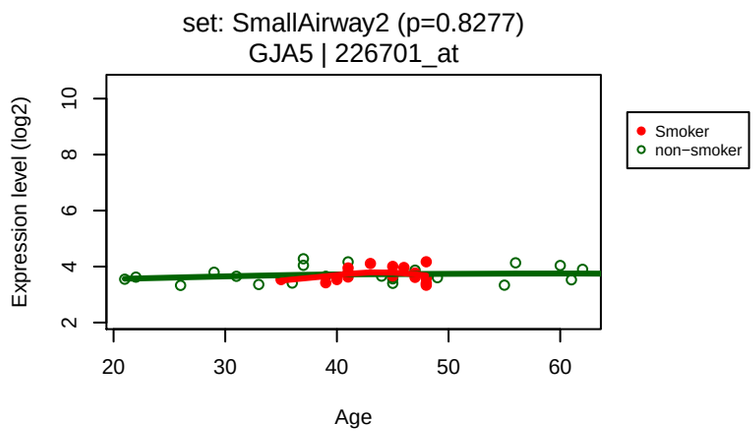
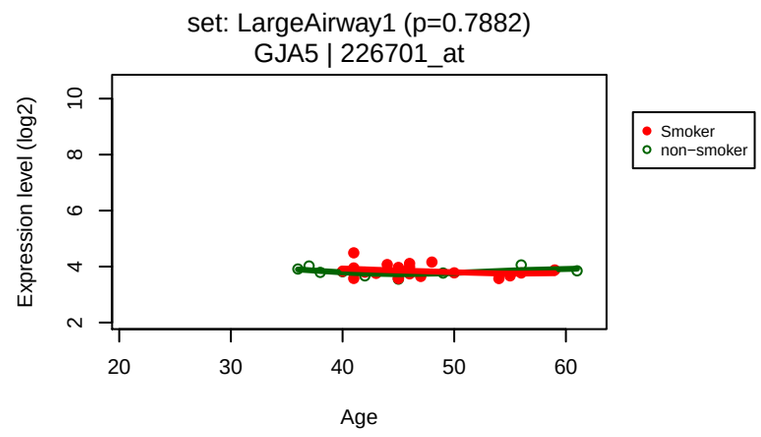

Gene expression of smokers and nonsmokers
gene: GJA5 | 226701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9266
p=0.8277
p=0.0889
p=0.7882
p=0.197
p=0.0218
p=0.5118
p=0.4738
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

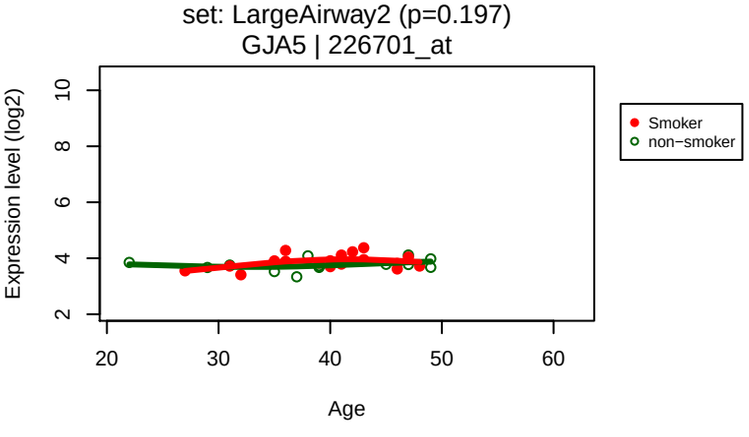

Gene expression ratio (smokers/nonsmokers)
gene: GJA5 | 226701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9266
p=0.8277
p=0.0889
p=0.7882
p=0.197
p=0.0218
p=0.5118
p=0.4738
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

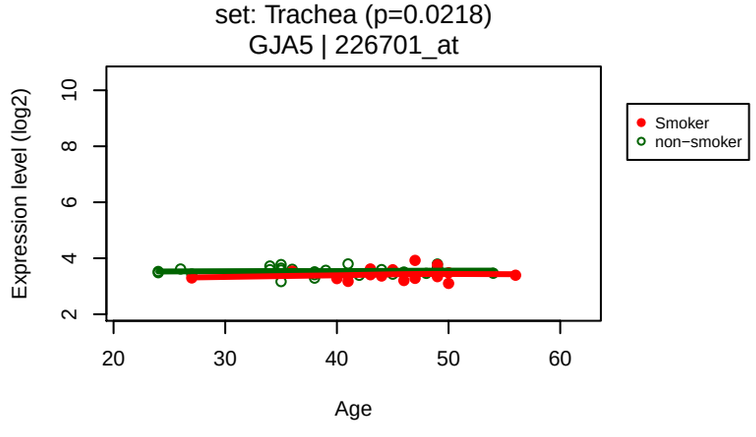

Gene expression ratio in Males
gene: GJA5 | 226701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9188
p=0.1895
p=0.0468
p=0.7238
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GJA5 | 226701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6556
p=0.8562
p=0.4322
p=0.3876
SmallAirway2
LargeAirway2
Trachea
Macrophage2

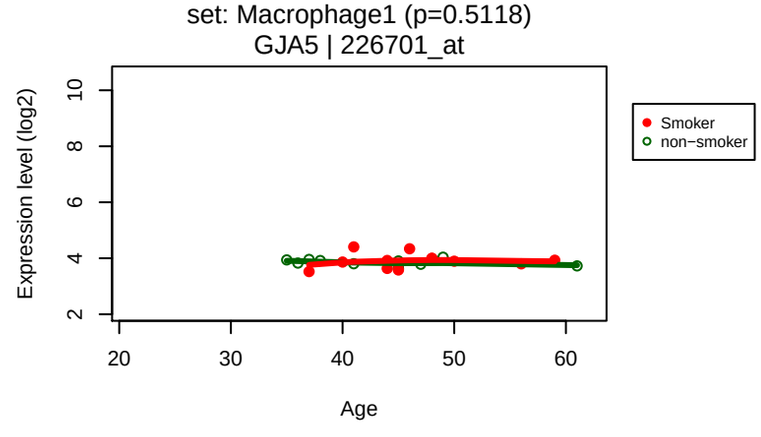

set: SmallAirway1 (p=0.9266)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker

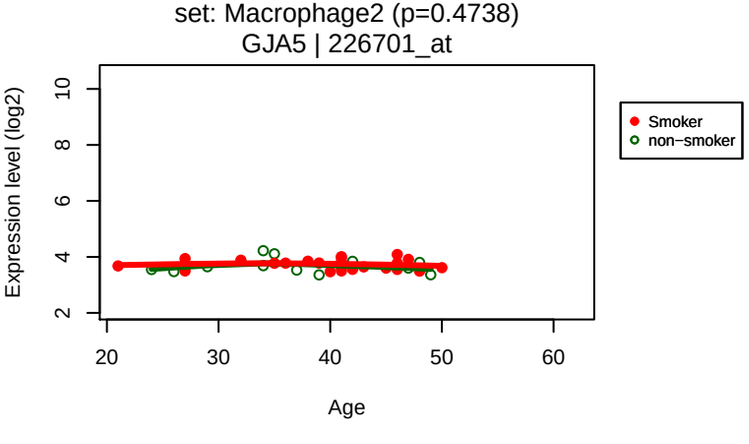

set: SmallAirway2 (p=0.8277)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker

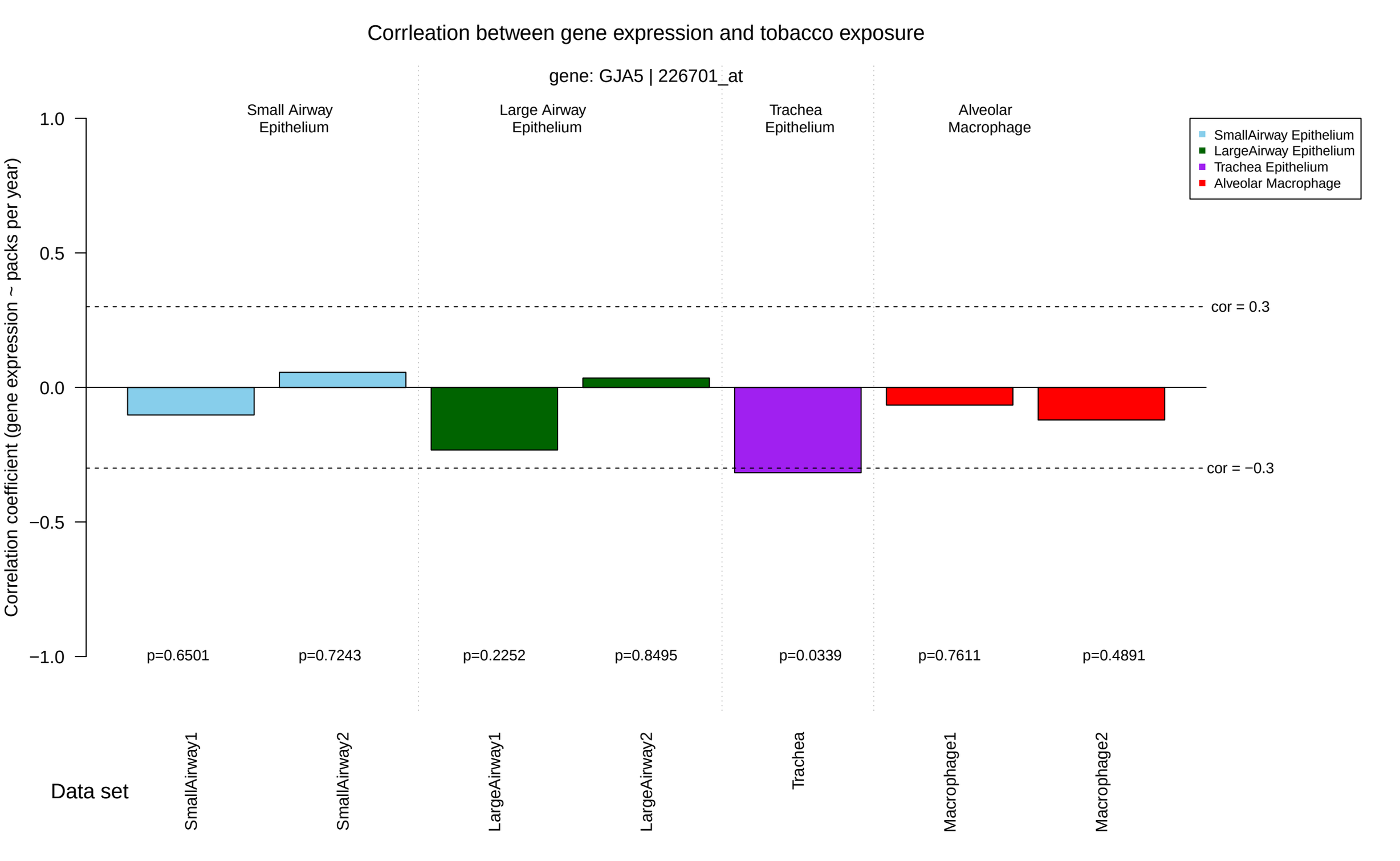

set: LargeAirway1 (p=0.7882)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.197)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0218)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5118)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4738)
GJA5 | 226701_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GJA5 | 226701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6501
p=0.7243
p=0.2252
p=0.8495
p=0.0339
p=0.7611
p=0.4891
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2